

Generative Artificial Intelligence

100 Questions: GenAI, LLMs, Transformers, Word2Vec, RNNs, LSTM

GARP RAI Exam Preparation

Q1: Generative AI Definition - Tutorial

QUESTION: What is Generative Artificial Intelligence (GenAI)?

OPTIONS:

- ❶ A branch of AI that analyzes existing data to make predictions
- ❷ A branch of AI that employs models to generate text, sound, or videos
- ❸ A branch of AI that only classifies images
- ❹ A branch of AI that performs regression analysis

ANSWER: B

DETAILED EXPLANATION:

- **Definition:** GenAI creates new content rather than just analyzing existing data
- **Key characteristic:** Generates text, sound, or videos
- **Example:** ChatGPT generates text, DALL-E generates images

WHY OTHER OPTIONS ARE WRONG:

- **A:** This describes traditional AI/machine learning - it analyzes, not generates
- **C:** This is too narrow - GenAI creates more than just images
- **D:** Regression is a traditional ML technique, not GenAI

EXAM TRAP: GenAI is about **creating** new content, not just analyzing!

QUESTION: What was significant about ChatGPT's release in November 2022?

OPTIONS:

- ❶ It was the first AI tool ever created
- ❷ It reached 1 million users in just 5 days, making it the fastest AI tool to do so
- ❸ It was the first chatbot
- ❹ It had no impact on AI adoption

ANSWER: B

DETAILED EXPLANATION:

- ChatGPT reached 1 million users in just 5 days
- Facebook took almost a year to reach this milestone
- Demonstrated rapid adoption of GenAI
- Sparked widespread interest and investment

WHY OTHER OPTIONS ARE WRONG:

- **A:** AI tools existed long before ChatGPT
- **C:** Chatbots existed before ChatGPT (ELIZA in 1960s)
- **D:** ChatGPT had enormous impact on AI adoption

EXAM TRAP: ChatGPT was the **fastest AI tool** to reach 1 million users!

Q3: GenAI vs Traditional AI - Tutorial

QUESTION: What is the key difference between Generative AI and Traditional AI?

OPTIONS:

- ❶ Traditional AI creates new content; GenAI analyzes existing data
- ❷ GenAI creates new content; Traditional AI analyzes existing data and makes predictions
- ❸ Both create new content equally
- ❹ Neither creates new content

ANSWER: B

DETAILED EXPLANATION:

- **Traditional AI:** Analyzes existing data, makes predictions/classifications
- **GenAI:** Creates new content (text, images, audio, video)
- **Traditional AI outputs:** Labels or numerical values
- **GenAI outputs:** Novel, creative content

WHY OTHER OPTIONS ARE WRONG:

- **A:** This reverses the definitions
- **C:** They are fundamentally different in output
- **D:** GenAI definitely creates new content

EXAM TRAP: GenAI = [creates](#); Traditional AI = [analyzes](#)!

Q4: GenAI Content Modalities - Tutorial

QUESTION: What are the five major content modalities in GenAI?

OPTIONS:

- ❶ Text, Image, Audio, Video, and Multimodal
- ❷ Text, Image, Sound, Animation, and 3D
- ❸ Numbers, Text, Images, Audio, and Video
- ❹ Text, Photos, Music, Movies, and Art

ANSWER: A

DETAILED EXPLANATION:

- **Text:** ChatGPT, Claude, Gemini
- **Image:** DALL-E, Stable Diffusion
- **Audio:** Text-to-speech, music generation
- **Video:** Sora, RunwayML
- **Multimodal:** Gemini, GPT-4 (multiple input/output types)

WHY OTHER OPTIONS ARE WRONG:

- **B:** "Animation" and "3D" are subcategories, not major modalities
- **C:** "Numbers" is traditional data, not a GenAI modality
- **D:** Uses casual terms, not the technical classification

EXAM TRAP: The five modalities are **Text, Image, Audio, Video, Multimodal!**

QUESTION: What does a multimodal GenAI model do?

OPTIONS:

- ❶ Only processes text inputs
- ❷ Takes multiple input types and generates multiple output types
- ❸ Only generates images
- ❹ Only processes a single type of data

ANSWER: B

DETAILED EXPLANATION:

- Multimodal models accept various input types
- Can generate output in different formats
- Example: Gemini can take text and images as input
- Example: GPT-4 can analyze images and generate text

WHY OTHER OPTIONS ARE WRONG:

- **A:** This describes text-only models (unimodal)
- **C:** This describes image-only models
- **D:** This describes unimodal models, not multimodal

EXAM TRAP: Multimodal = **multiple input and output types!**

QUESTION: What is a Large Language Model (LLM)?

OPTIONS:

- ❶ A small neural network with few parameters
- ❷ A large neural network with billions of parameters, trained on vast text data
- ❸ A model that only generates images
- ❹ A model that only processes numerical data

ANSWER: B

DETAILED EXPLANATION:

- LLMs have billions to trillions of parameters
- Trained on very large text corpora
- Use transformer architecture
- Examples: GPT-3 (175B parameters), GPT-4

WHY OTHER OPTIONS ARE WRONG:

- **A:** LLMs are LARGE, not small
- **C:** LLMs are text-focused, not image generation
- **D:** LLMs process text, not primarily numerical data

EXAM TRAP: LLMs are **large neural networks** with billions of parameters!

QUESTION: What is the key innovation of transformers in NLP?

OPTIONS:

- ❶ They use only feedforward networks
- ❷ They use the "attention" mechanism to understand word relationships
- ❸ They are slower than RNNs
- ❹ They process text sequentially

ANSWER: B

DETAILED EXPLANATION:

- Transformers use self-attention mechanism
- Attention allows understanding relationships between all words
- Process entire sequence in parallel
- Foundation for modern LLMs

WHY OTHER OPTIONS ARE WRONG:

- **A:** Transformers use attention AND feedforward networks
- **C:** Transformers are faster than RNNs (parallel processing)
- **D:** Transformers process in parallel, not sequentially

EXAM TRAP: Transformers = [attention mechanism](#)!

QUESTION: What does DALL-E do?

OPTIONS:

- ❶ Generates text from images
- ❷ Creates images from text prompts
- ❸ Generates audio from text
- ❹ Analyzes financial data

ANSWER: B

DETAILED EXPLANATION:

- DALL-E is developed by OpenAI
- Takes text prompts and creates images
- Can generate novel, creative images
- Powerful image generation tool

WHY OTHER OPTIONS ARE WRONG:

- **A:** This describes image-to-text (opposite of DALL-E)
- **C:** Audio generation is a different modality
- **D:** Financial analysis is not DALL-E's function

EXAM TRAP: DALL-E = image generation from text prompts!

QUESTION: What are the two competing networks in a Generative Adversarial Network (GAN)?

OPTIONS:

- ❶ Encoder and Decoder
- ❷ Generator and Discriminator
- ❸ Input and Output
- ❹ Training and Testing

ANSWER: B

DETAILED EXPLANATION:

- **Generator:** Creates simulated data, learns to fool discriminator
- **Discriminator:** Identifies real vs fake data, gives feedback
- Process iterates until generated data is indistinguishable
- Used for images, audio, video creation

WHY OTHER OPTIONS ARE WRONG:

- **A:** Encoder/Decoder are in VAEs, not GANs
- **C:** Input/Output are general terms, not GAN-specific
- **D:** Training/Testing are ML phases, not GAN architecture

EXAM TRAP: GANs = **Generator + Discriminator** competing!

Q10: GenAI vs LLM Relationship - Tutorial

QUESTION: Which statement correctly describes the relationship between GenAI and LLMs?

OPTIONS:

- ❶ GenAI and LLMs are the same thing
- ❷ LLMs are a subset of GenAI that focus on text generation
- ❸ GenAI is a subset of LLMs
- ❹ They are completely unrelated

ANSWER: B

DETAILED EXPLANATION:

- GenAI is broader - includes images, audio, video
- LLMs specifically focus on text generation
- All LLMs are GenAI, but not all GenAI are LLMs
- GenAI = umbrella term; LLMs = specific type

WHY OTHER OPTIONS ARE WRONG:

- **A:** GenAI is broader than just LLMs
- **C:** This reverses the relationship
- **D:** They are related - LLMs are a type of GenAI

EXAM TRAP: All LLMs are GenAI, but **not all GenAI are LLMs!**

QUESTION: What is a major limitation of the Bag of Words (BoW) approach?

OPTIONS:

- ❶ It captures semantic meaning of words
- ❷ It creates dense vectors with few zeros
- ❸ It treats words independently and loses context
- ❹ It handles negation words perfectly

ANSWER: C

DETAILED EXPLANATION:

- ❶ BoW ignores word order and context
- ❷ "not good" and "good" both contain "good"
- ❸ Cannot identify similar meanings ("amazing" vs "fantastic")
- ❹ Creates sparse vectors with many zeros

WHY OTHER OPTIONS ARE WRONG:

- ❶ **A:** BoW does NOT capture semantic meaning
- ❷ **B:** BoW creates SPARSE, not dense vectors
- ❹ **D:** BoW struggles with negation words

EXAM TRAP: BoW loses context and semantic meaning!

QUESTION: Who created Word2Vec and when?

OPTIONS:

- ❶ OpenAI in 2020
- ❷ Google team led by Tomas Mikolov in 2013
- ❸ Facebook in 2018
- ❹ Microsoft in 2015

ANSWER: B

DETAILED EXPLANATION:

- Word2Vec created by Google team (2013)
- Led by Tomas Mikolov
- Alternative to conventional BoW
- Revolutionary for NLP

WHY OTHER OPTIONS ARE WRONG:

- **A:** OpenAI was not involved in Word2Vec
- **C:** Facebook created other models (BART, RoBERTa)
- **D:** Microsoft was not involved in Word2Vec

EXAM TRAP: Word2Vec = **Google 2013** by Tomas Mikolov!

Q13: Word Embeddings Definition - Tutorial

QUESTION: What are word embeddings in NLP?

OPTIONS:

- ❶ Sparse vectors with mostly zeros
- ❷ Dense vector representations of words that encode semantic information
- ❸ A list of synonyms for each word
- ❹ A method for removing stop words

ANSWER: B

DETAILED EXPLANATION:

- Dense vectors (fewer dimensions)
- Similar words have similar vectors
- Encode semantic meaning
- Generated through neural network training

WHY OTHER OPTIONS ARE WRONG:

- **A:** Embeddings are DENSE, not sparse
- **C:** Embeddings are vectors, not word lists
- **D:** Embeddings are for representation, not removal

EXAM TRAP: Embeddings = dense vectors capturing meaning!

QUESTION: What is the Continuous Bag of Words (CBoW) architecture in Word2Vec?

OPTIONS:

- ❶ Predicts context words from a target word
- ❷ Predicts a target word from surrounding context words
- ❸ Predicts the next word in a sequence
- ❹ Only analyzes single words

ANSWER: B

DETAILED EXPLANATION:

- "Fill in the blank" technique
- Input: context words (before and after)
- Output: target word (the missing word)
- Goal: $P(w_i | w_{i-c}, \dots, w_{i+c})$

WHY OTHER OPTIONS ARE WRONG:

- **A:** This describes Skip-gram (context from target)
- **C:** This describes autoregressive models
- **D:** CBoW uses context, not just single words

EXAM TRAP: CBoW = context → target word!

QUESTION: What is the Skip-gram architecture in Word2Vec?

OPTIONS:

- ❶ Predicts a target word from surrounding context words
- ❷ Predicts surrounding context words from a target word
- ❸ Predicts the next sentence
- ❹ Analyzes word frequency only

ANSWER: B

DETAILED EXPLANATION:

- "Word association" technique
- Input: target word
- Output: context words (surrounding words)
- Goal: $P(w_{i-c}, \dots, w_{i+c} | w_i)$

WHY OTHER OPTIONS ARE WRONG:

- **A:** This describes CBoW
- **C:** Skip-gram predicts words, not sentences
- **D:** Skip-gram uses neural networks, not just frequency

EXAM TRAP: Skip-gram = **target** → **context** words!

Q16: CBoW vs Skip-gram - Tutorial

QUESTION: What is the key difference between CBoW and Skip-gram?

OPTIONS:

- ❶ CBoW is for English; Skip-gram for other languages
- ❷ CBoW predicts target from context; Skip-gram predicts context from target
- ❸ CBoW is slower than Skip-gram
- ❹ There is no difference - they are the same

ANSWER: B

DETAILED EXPLANATION:

- **CBoW:** context $\rightarrow$ target
- **Skip-gram:** target $\rightarrow$ context
- Both are Word2Vec architectures
- Skip-gram better for rare words; CBoW faster

WHY OTHER OPTIONS ARE WRONG:

- **A:** Both work for all languages
- **C:** CBoW is actually faster
- **D:** They are distinctly different architectures

EXAM TRAP: CBoW = context $\rightarrow$ target; Skip-gram = target $\rightarrow$ context!

QUESTION: How does Word2Vec reduce dimensionality compared to BoW?

OPTIONS:

- ❶ Word2Vec uses more dimensions than BoW
- ❷ Word2Vec uses dense 300-dimensional vectors instead of vocabulary-sized sparse vectors
- ❸ Word2Vec doesn't reduce dimensionality
- ❹ Word2Vec uses one-hot encoding

ANSWER: B

DETAILED EXPLANATION:

- Traditional BoW: vocabulary size dimensions (e.g., 100,000)
- Word2Vec: typically 300 dimensions
- Reduction: 99.7% fewer weights!
- Dense vs sparse representation

WHY OTHER OPTIONS ARE WRONG:

- **A:** Word2Vec uses FEWER dimensions
- **C:** Dimensionality reduction is a key feature
- **D:** Word2Vec uses embeddings, not one-hot encoding

EXAM TRAP: Word2Vec reduces dimensions from **vocabulary size** to **300**!

Q18: Cosine Similarity Formula - Tutorial

QUESTION: What is the formula for cosine similarity between two vectors P and Q?

OPTIONS:

❶ $\cos(\theta) = \frac{P+Q}{||P|| \times ||Q||}$

❷ $\cos(\theta) = \frac{P \cdot Q}{||P|| \times ||Q||}$

❸ $\cos(\theta) = \frac{||P||}{||Q||}$

❹ $\cos(\theta) = P \times Q$

ANSWER: B

DETAILED EXPLANATION:

- $P \cdot Q$ = dot product
- $||P||$ = magnitude (norm) of vector P
- Range: $[-1, 1]$
- 1 = identical, 0 = orthogonal, -1 = opposite

WHY OTHER OPTIONS ARE WRONG:

- **A:** Addition is not the correct operation
- **C:** Only compares magnitudes, ignores direction
- **D:** Missing normalization (dividing by norms)

EXAM TRAP: Cosine similarity = **dot product** / (**norm1** × **norm2**)!

QUESTION: What does a cosine similarity of 0.95 indicate?

OPTIONS:

- ❶ The words are completely unrelated
- ❷ The words are very similar (close in embedding space)
- ❸ The words are opposites
- ❹ The words are identical

ANSWER: B

DETAILED EXPLANATION:

- Range: -1 to 1
- 0.95 is very close to 1.0
- Indicates high similarity
- Words are likely similar in meaning

WHY OTHER OPTIONS ARE WRONG:

- **A:** 0.95 is HIGH, not unrelated (0)
- **C:** Opposites would be close to -1
- **D:** Identical would be exactly 1.0

EXAM TRAP: Higher cosine similarity = **more similar words!**

QUESTION: What is the result of the operation "King - Man + Woman" in word embeddings?

OPTIONS:

- 1 Prince
- 2 Queen
- 3 Royalty
- 4 Knight

ANSWER: B

DETAILED EXPLANATION:

- "King - Man + Woman $\approx$ Queen"
- Captures gender relationship
- Vector arithmetic preserves semantic relationships
- Famous Word2Vec analogy

WHY OTHER OPTIONS ARE WRONG:

- **A:** Prince is a different royal title
- **C:** Royalty is too general
- **D:** Knight is not a royal title

EXAM TRAP: "King - Man + Woman" = Queen!

QUESTION: What is a key limitation of word embeddings like Word2Vec?

OPTIONS:

- ❶ They capture sentence-level context perfectly
- ❷ They cannot capture context-dependent word meanings
- ❸ They always require a fixed context window
- ❹ Both B and C are limitations

ANSWER: D

DETAILED EXPLANATION:

- Same word has same embedding regardless of context
- "bat" (animal) vs "bat" (sports) - same embedding
- Fixed context window must be specified
- Cannot handle distant relationships beyond window

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is false - they DO NOT capture sentence-level context
- **B:** This is true but incomplete
- **C:** This is true but incomplete

EXAM TRAP: Word2Vec gives **same vector for different contexts!**

QUESTION: Which of the following is an alternative to Word2Vec for word embeddings?

OPTIONS:

- ❶ GloVe (Global Vectors)
- ❷ BERT (Bidirectional Encoder Representations)
- ❸ Both GloVe and BERT
- ❹ None of the above

ANSWER: C

DETAILED EXPLANATION:

- **GloVe:** Stanford University, combines matrix factorization
- **BERT:** Contextual embeddings (different vectors for different contexts)
- Both are alternatives/improvements over Word2Vec

WHY OTHER OPTIONS ARE WRONG:

- **A:** GloVe is correct but incomplete
- **B:** BERT is correct but incomplete
- **D:** Both A and B are valid alternatives

EXAM TRAP: GloVe and BERT are **both alternatives** to Word2Vec!

QUESTION: What is t-SNE used for in the context of word embeddings?

OPTIONS:

- ❶ To create word embeddings
- ❷ To visualize high-dimensional embeddings in 2D
- ❸ To calculate cosine similarity
- ❹ To train neural networks

ANSWER: B

DETAILED EXPLANATION:

- t-SNE = t-distributed Stochastic Neighbor Embedding
- Reduces high-dimensional embeddings to 2D
- Visualizes relationships between words
- Clusters show semantic groups

WHY OTHER OPTIONS ARE WRONG:

- **A:** t-SNE visualizes, doesn't create embeddings
- **C:** Cosine similarity is a separate measure
- **D:** t-SNE is for visualization, not training

EXAM TRAP: t-SNE is for **visualization**, not creation!

QUESTION: What is a common application of word embeddings?

OPTIONS:

- ❶ Sentiment analysis
- ❷ Document clustering
- ❸ Input to transformer models
- ❹ All of the above

ANSWER: D

DETAILED EXPLANATION:

- Clustering: group similar documents
- Sentiment analysis: understand word meanings
- Transformer input: pre-trained embeddings
- Foundation for modern NLP systems

WHY OTHER OPTIONS ARE WRONG:

- **A:** Correct but incomplete
- **B:** Correct but incomplete
- **C:** Correct but incomplete

EXAM TRAP: Embeddings are **foundational** for many NLP tasks!

QUESTION: What is SBert used for in NLP?

OPTIONS:

- ❶ Word-level embeddings
- ❷ Sentence-level embeddings
- ❸ Document-level embeddings
- ❹ Image generation

ANSWER: B

DETAILED EXPLANATION:

- SBert = Sentence BERT
- Captures sentence-level meaning
- Better for sentence comparison
- Beyond word-level embeddings

WHY OTHER OPTIONS ARE WRONG:

- **A:** Word embeddings are Word2Vec/GloVe
- **C:** Doc2Vec is for document embeddings
- **D:** SBert is text-based, not image generation

EXAM TRAP: SBert = sentence-level embeddings!

QUESTION: What is a Recurrent Neural Network (RNN)?

OPTIONS:

- ❶ A feedforward neural network with no loops
- ❷ A neural network designed for sequential data with memory
- ❸ A network that only processes images
- ❹ A network that has no hidden layers

ANSWER: B

DETAILED EXPLANATION:

- Has memory (hidden cell)
- Value at time t depends on previous values
- Designed for sequential data (text, speech, time series)
- Outputs become inputs for next step

WHY OTHER OPTIONS ARE WRONG:

- **A:** FNNs have no loops; RNNs have feedback loops
- **C:** RNNs handle sequential data, not just images
- **D:** RNNs have hidden layers (that's their memory)

EXAM TRAP: RNNs have **memory** for sequential data!

QUESTION: What is the key difference between RNNs and Feedforward Neural Networks (FNNs)?

OPTIONS:

- ❶ FNNs have memory; RNNs do not
- ❷ RNNs have memory (hidden state); FNNs do not
- ❸ Both have memory
- ❹ Both are identical

ANSWER: B

DETAILED EXPLANATION:

- **FNN:** $y = f(x)$ - no memory
- **RNN:** $y_t = f(x_t, h_{t-1})$ - has memory
- RNNs process sequential data
- FNNs process static data

WHY OTHER OPTIONS ARE WRONG:

- **A:** This reverses the definitions
- **C:** FNNs do NOT have memory
- **D:** They are fundamentally different

EXAM TRAP: RNNs have **memory**; FNNs do not!

QUESTION: What is the hidden state h_t in an RNN?

OPTIONS:

- ❶ The output at time t only
- ❷ The memory/state at time t that depends on previous states
- ❸ The input at time t
- ❹ The final output of the network

ANSWER: B

DETAILED EXPLANATION:

- $h_t = f(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$
- Depends on current input AND previous state
- Stores information from previous time steps
- Represents the network's memory

WHY OTHER OPTIONS ARE WRONG:

- **A:** Hidden state is not just the output
- **C:** Input is x_t , not hidden state
- **D:** Hidden state is intermediate, not final

EXAM TRAP: Hidden state = **memory at time t!**

Q29: Vanishing Gradient Problem - Tutorial

QUESTION: What is the vanishing gradient problem in RNNs?

OPTIONS:

- ❶ Gradients become very large during training
- ❷ Gradients become very small during backpropagation, limiting long-range learning
- ❸ Gradients oscillate randomly
- ❹ Gradients remain constant throughout training

ANSWER: B

DETAILED EXPLANATION:

- Gradients multiplied through many time steps
- Each multiplication by < 1
- Product approaches zero
- Long-range dependencies are lost

WHY OTHER OPTIONS ARE WRONG:

- **A:** This describes exploding gradients
- **C:** This describes oscillations
- **D:** Gradients change during training

EXAM TRAP: Vanishing gradients = **very small gradients** in early layers!

QUESTION: What was LSTM designed to address?

OPTIONS:

- ❶ The need for faster training
- ❷ The vanishing gradient problem in RNNs
- ❸ The need for larger datasets
- ❹ The need for simpler architectures

ANSWER: B

DETAILED EXPLANATION:

- LSTM = Long Short-Term Memory
- Specifically designed to address vanishing gradients
- Uses gates to regulate information flow
- Can capture long-range dependencies

WHY OTHER OPTIONS ARE WRONG:

- **A:** LSTMs are more complex, not faster
- **C:** LSTMs don't address data size issues
- **D:** LSTMs are more complex, not simpler

EXAM TRAP: LSTM = [solution to vanishing gradient](#) problem!

QUESTION: What are the three gates in a basic LSTM cell?

OPTIONS:

- ❶ Input, Output, and Hidden
- ❷ Forget, Input, and Output
- ❸ Memory, Forget, and Output
- ❹ Encoder, Decoder, and Memory

ANSWER: B

DETAILED EXPLANATION:

- **Forget Gate:** What to discard from memory
- **Input Gate:** What new information to store
- **Output Gate:** What to output
- These gates regulate information flow

WHY OTHER OPTIONS ARE WRONG:

- **A:** "Hidden" is not a gate
- **C:** "Memory" is not a gate
- **D:** Encoder/Decoder are for transformers, not LSTMs

EXAM TRAP: LSTM gates = Forget, Input, Output!

Q32: LSTM vs RNN Comparison - Tutorial

QUESTION: Which statement correctly compares LSTM and RNN?

OPTIONS:

- ❶ RNN is more powerful than LSTM
- ❷ LSTM handles long-term dependencies better than RNN
- ❸ RNN and LSTM are identical
- ❹ LSTM is simpler than RNN

ANSWER: B

DETAILED EXPLANATION:

- RNN struggles with long-term dependencies (vanishing gradients)
- LSTM solves this with gates
- LSTM is more powerful but more complex
- LSTM can capture relationships over longer distances

WHY OTHER OPTIONS ARE WRONG:

- **A:** LSTM is more powerful than RNN
- **C:** They are different architectures
- **D:** LSTM is more complex, not simpler

EXAM TRAP: LSTM **outperforms** RNN for long sequences!

QUESTION: What is the purpose of the Forget Gate in an LSTM?

OPTIONS:

- ❶ To determine what new information to store
- ❷ To determine what information to discard from memory
- ❸ To determine what to output
- ❹ To initialize the network

ANSWER: B

DETAILED EXPLANATION:

- $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$
- Decides which information is no longer relevant
- Prevents memory from overflowing
- Essential for long-term learning

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is the Input Gate's function
- **C:** This is the Output Gate's function
- **D:** This is initialization, not a gate function

EXAM TRAP: Forget Gate = what to discard from memory!

QUESTION: What is a common application of RNNs in finance?

OPTIONS:

- ❶ Image classification
- ❷ Time series analysis and prediction
- ❸ Image generation
- ❹ Image segmentation

ANSWER: B

DETAILED EXPLANATION:

- RNNs excel at sequential data
- Stock price prediction
- Volatility forecasting
- Economic indicator forecasting

WHY OTHER OPTIONS ARE WRONG:

- **A:** CNNs are better for image classification
- **C:** GANs/Diffusion models are for image generation
- **D:** CNNs are used for image segmentation

EXAM TRAP: RNNs = time series and sequential data!

QUESTION: What is a major limitation of RNNs that transformers address?

OPTIONS:

- ❶ RNNs are too fast
- ❷ RNNs cannot parallelize effectively
- ❸ RNNs require too little data
- ❹ RNNs are too interpretable

ANSWER: B

DETAILED EXPLANATION:

- RNNs process sequentially (word by word)
- Cannot parallelize across sequence
- Transformers process entire sequence in parallel
- Transformers are much faster

WHY OTHER OPTIONS ARE WRONG:

- **A:** RNNs are slow, not too fast
- **C:** RNNs require lots of data
- **D:** RNNs are harder to interpret than feedforward, not more

EXAM TRAP: RNNs cannot **parallelize**; transformers can!

QUESTION: What paper introduced the transformer architecture?

OPTIONS:

- ❶ "Attention is All You Need" (2017)
- ❷ "BERT: Pre-training of Deep Bidirectional Transformers" (2018)
- ❸ "Generative Pre-trained Transformer" (2018)
- ❹ "Language Models are Few-Shot Learners" (2020)

ANSWER: A

DETAILED EXPLANATION:

- "Attention is All You Need" by Vaswani et al. (2017)
- Introduced the transformer architecture
- Revolutionized NLP
- Foundation for all modern LLMs

WHY OTHER OPTIONS ARE WRONG:

- **B:** BERT came after transformers
- **C:** GPT came after transformers
- **D:** GPT-3 paper came after transformers

EXAM TRAP: Transformers introduced in "Attention is All You Need" (2017)!

QUESTION: What are the two main components of a transformer?

OPTIONS:

- ❶ Generator and Discriminator
- ❷ Encoder and Decoder
- ❸ Input and Output layers only
- ❹ Attention and Recurrent layers

ANSWER: B

DETAILED EXPLANATION:

- **Encoder:** Processes input, converts to vectors, "understands" text
- **Decoder:** Generates output based on input/prompt
- Some models use only encoder (BERT) or only decoder (GPT)
- Original had both (encoder-decoder)

WHY OTHER OPTIONS ARE WRONG:

- **A:** These are GAN components, not transformer
- **C:** Transformers have more than just input/output
- **D:** Transformers use attention, not recurrence

EXAM TRAP: Transformer = Encoder + Decoder!

QUESTION: What is self-attention in a transformer?

OPTIONS:

- ❶ Ignoring all other words in a sentence
- ❷ Calculating relationships between all words in a sequence
- ❸ Only paying attention to the first word
- ❹ Paying attention only to nouns

ANSWER: B

DETAILED EXPLANATION:

- Each word analyzes all other words
- Captures relationships between all words
- Creates semantic understanding
- Enables long-range dependencies

WHY OTHER OPTIONS ARE WRONG:

- **A:** This would be no-attention
- **C:** This ignores important context
- **D:** Attention applies to ALL words

EXAM TRAP: Self-attention = all words analyze all other words!

QUESTION: What are the three roles in the attention mechanism?

OPTIONS:

- ❶ Input, Output, and Hidden
- ❷ Query, Key, and Value
- ❸ Encoder, Decoder, and Memory
- ❹ Training, Validation, and Testing

ANSWER: B

DETAILED EXPLANATION:

- **Query (Q):** Current token asking "what should I attend to?"
- **Key (K):** Other tokens responding to the query
- **Value (V):** Content of tokens used to create context
- All three are derived from input embeddings

WHY OTHER OPTIONS ARE WRONG:

- **A:** These are general terms, not attention-specific
- **C:** Encoder/Decoder are transformer components
- **D:** These are ML phases, not attention roles

EXAM TRAP: Attention uses **Query, Key, Value** vectors!

Q40: Attention Score Calculation - Tutorial

QUESTION: How is the attention score between query q_i and key k_j calculated?

OPTIONS:

- ❶ $score(q_i, k_j) = q_i + k_j$
- ❷ $score(q_i, k_j) = q_i \cdot k_j$
- ❸ $score(q_i, k_j) = q_i - k_j$
- ❹ $score(q_i, k_j) = q_i / k_j$

ANSWER: B

DETAILED EXPLANATION:

- Dot product measures similarity
- Higher dot product = more relevant
- Then normalized with softmax
- Weighted sum of value vectors

WHY OTHER OPTIONS ARE WRONG:

- **A:** Addition doesn't measure similarity
- **C:** Subtraction doesn't measure similarity
- **D:** Division doesn't measure similarity

EXAM TRAP: Attention score = dot product of Query and Key!

QUESTION: What is the purpose of multi-head attention?

OPTIONS:

- ❶ To reduce computation
- ❷ To capture different types of relationships in parallel
- ❸ To make the model simpler
- ❹ To eliminate the need for feedforward layers

ANSWER: B

DETAILED EXPLANATION:

- Multiple attention heads in parallel
- Each head focuses on different aspects
- Captures different relationships
- Example: GPT has 12 heads

WHY OTHER OPTIONS ARE WRONG:

- **A:** Multi-head increases computation
- **C:** Multi-head makes models more complex
- **D:** Feedforward layers are still needed

EXAM TRAP: Multi-head captures **diverse relationships!**

QUESTION: Why are positional embeddings needed in transformers?

OPTIONS:

- ❶ Transformers process words in order, so they don't need positions
- ❷ Transformers process words in parallel, so they need position information
- ❸ Positional embeddings are optional
- ❹ Positional embeddings are only for RNNs

ANSWER: B

DETAILED EXPLANATION:

- Transformers process all words simultaneously
- Without positional info, word order is lost
- Positional embeddings encode position
- "dog bites man" vs "man bites dog"

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is incorrect - positions are needed
- **C:** Positional embeddings are necessary
- **D:** Positional embeddings are for transformers

EXAM TRAP: Positional embeddings = **needed** because transformers process in parallel!

Q43: Transformer Advantage - Tutorial

QUESTION: What is a key advantage of transformers over RNNs?

OPTIONS:

- ❶ Transformers are always more interpretable
- ❷ Transformers process in parallel, making them faster
- ❸ Transformers require less data
- ❹ Transformers are simpler

ANSWER: B

DETAILED EXPLANATION:

- RNNs process sequentially (slow)
- Transformers process entire sequence at once (fast)
- Parallel processing = faster training
- Better hardware utilization

WHY OTHER OPTIONS ARE WRONG:

- **A:** Transformers are LESS interpretable than RNNs
- **C:** Transformers require MORE data
- **D:** Transformers are MORE complex

EXAM TRAP: Transformers = parallel processing = faster!

QUESTION: What type of architecture does BERT use?

OPTIONS:

- ❶ Encoder-only
- ❷ Decoder-only
- ❸ Encoder-Decoder
- ❹ Neither encoder nor decoder

ANSWER: A

DETAILED EXPLANATION:

- BERT = Bidirectional Encoder Representations
- Uses only the encoder part
- No decoder (not generative)
- Good for understanding tasks

WHY OTHER OPTIONS ARE WRONG:

- **B:** GPT uses decoder-only
- **C:** BART uses encoder-decoder
- **D:** BERT uses encoders

EXAM TRAP: BERT = **encoder-only** architecture!

QUESTION: How is BERT trained?

OPTIONS:

- ❶ Only using labeled data
- ❷ Using masked language modeling and next sentence prediction
- ❸ Using reinforcement learning
- ❹ Using only supervised learning

ANSWER: B

DETAILED EXPLANATION:

- **Masked Language Modeling:** 15% of words masked, model predicts them
- **Next Sentence Prediction:** Predict if sentences are sequential
- Self-supervised (labels come from data)
- Learns bidirectional context

WHY OTHER OPTIONS ARE WRONG:

- **A:** BERT uses unlabeled data for pre-training
- **C:** Reinforcement learning is for RLHF
- **D:** BERT uses self-supervised learning

EXAM TRAP: BERT = masked LM + next sentence prediction!

QUESTION: What type of architecture does GPT use?

OPTIONS:

- ❶ Encoder-only
- ❷ Decoder-only
- ❸ Encoder-Decoder
- ❹ No transformer architecture

ANSWER: B

DETAILED EXPLANATION:

- GPT = Generative Pre-trained Transformer
- Uses only the decoder part
- Autoregressive (predicts next token)
- Good for generation tasks

WHY OTHER OPTIONS ARE WRONG:

- **A:** BERT uses encoder-only
- **C:** BART uses encoder-decoder
- **D:** GPT uses transformer architecture

EXAM TRAP: GPT = **decoder-only** architecture!

QUESTION: What type of architecture does **BART** use?

OPTIONS:

- ❶ Encoder-only
- ❷ Decoder-only
- ❸ Encoder-Decoder
- ❹ No transformer architecture

ANSWER: C

DETAILED EXPLANATION:

- BART = Bidirectional and Auto-Regressive Transformer
- Uses both encoder and decoder
- Most versatile
- Good for many tasks (summarization, translation)

WHY OTHER OPTIONS ARE WRONG:

- **A:** BERT uses encoder-only
- **B:** GPT uses decoder-only
- **D:** BART uses transformer architecture

EXAM TRAP: BART = **encoder-decoder** architecture!

QUESTION: How does RoBERTa improve upon BERT?

OPTIONS:

- ❶ It uses fewer training cycles
- ❷ It uses different masking patterns and larger dataset
- ❸ It removes the encoder
- ❹ It uses only supervised learning

ANSWER: B

DETAILED EXPLANATION:

- RoBERTa = Robustly Optimized BERT Approach
- Different masking patterns each cycle
- Larger training dataset
- More training cycles
- Multi-language web crawled data

WHY OTHER OPTIONS ARE WRONG:

- **A:** RoBERTa uses MORE training cycles
- **C:** RoBERTa still uses encoders
- **D:** RoBERTa uses self-supervised learning

EXAM TRAP: RoBERTa = improved BERT with more data and better masking!

QUESTION: What is FinBERT specifically designed for?

OPTIONS:

- ❶ General purpose NLP
- ❷ Financial sentiment analysis
- ❸ Image generation
- ❹ Speech recognition

ANSWER: B

DETAILED EXPLANATION:

- FinBERT is a financial text specialization
- Trained on Reuters financial documents
- Corporate statements and analyst reports
- Optimal for financial sentiment analysis

WHY OTHER OPTIONS ARE WRONG:

- **A:** FinBERT is domain-specific, not general
- **C:** FinBERT is for text, not images
- **D:** FinBERT is for text analysis, not speech

EXAM TRAP: FinBERT = financial sentiment analysis!

QUESTION: What is a disadvantage of transformers?

OPTIONS:

- ❶ They are too interpretable
- ❷ They are computationally intensive
- ❸ They cannot handle long-range dependencies
- ❹ They process sequentially

ANSWER: B

DETAILED EXPLANATION:

- Transformers require powerful hardware
- More computationally intensive than RNNs
- Higher energy consumption
- Longer training times

WHY OTHER OPTIONS ARE WRONG:

- **A:** Transformers are LESS interpretable
- **C:** Transformers EXCEL at long-range dependencies
- **D:** Transformers process in parallel, not sequentially

EXAM TRAP: Transformers are **resource-intensive**!

QUESTION: Approximately how many parameters does GPT-3 have?

OPTIONS:

- ❶ 100 million
- ❷ 175 billion
- ❸ 1 trillion
- ❹ 1 million

ANSWER: B

DETAILED EXPLANATION:

- GPT-3 has approximately 175 billion parameters
- Much larger than GPT-1 (100 million)
- Enables generative capacity
- Trained on vast web text

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is GPT-1's size
- **C:** Some newer models approach this, but not GPT-3
- **D:** This is far too small

EXAM TRAP: GPT-3 = 175 billion parameters!

QUESTION: What are the three types of LLMs based on architecture?

OPTIONS:

- ❶ Autoencoding, Autoregressive, and Combination
- ❷ Encoder, Decoder, and Memory
- ❸ Training, Validation, and Testing
- ❹ Supervised, Unsupervised, and Reinforcement

ANSWER: A

DETAILED EXPLANATION:

- **Autoencoding:** BERT (predicts masked words)
- **Autoregressive:** GPT (predicts next word)
- **Combination:** BART (both encoder and decoder)

WHY OTHER OPTIONS ARE WRONG:

- **B:** These are components, not LLM types
- **C:** These are ML phases, not LLM types
- **D:** These are learning paradigms, not LLM types

EXAM TRAP: LLM types = Autoencoding, Autoregressive, Combination!

QUESTION: What is the first step in LLM training?

OPTIONS:

- ❶ Fine-tuning
- ❷ Data preparation (cleaning, filtering, balancing)
- ❸ Decoding
- ❹ Reinforcement learning

ANSWER: B

DETAILED EXPLANATION:

- Clean training data (corpus)
- Filter and remove duplicates
- Remove noisy data and punctuation
- Balance class distributions
- Foundation for all subsequent steps

WHY OTHER OPTIONS ARE WRONG:

- **A:** Fine-tuning happens after pre-training
- **C:** Decoding happens at the end
- **D:** RLHF happens during alignment

EXAM TRAP: First step = data preparation!

QUESTION: What is tokenization in LLM training?

OPTIONS:

- ❶ Removing all punctuation
- ❷ Separating data into small parts (tokens) for efficient processing
- ❸ Converting text to lowercase
- ❹ Removing stop words

ANSWER: B

DETAILED EXPLANATION:

- Split text into tokens (words or subwords)
- Enables efficient processing
- Store positional information
- Foundation for model understanding

WHY OTHER OPTIONS ARE WRONG:

- **A:** Tokenization is more than removing punctuation
- **C:** Lowercasing is a pre-processing step
- **D:** Stop word removal is a pre-processing step

EXAM TRAP: Tokenization = **splitting into tokens!**

QUESTION: What happens during LLM pre-training?

OPTIONS:

- ❶ The model is fine-tuned on labeled data
- ❷ The model is trained on unlabeled text to understand word relationships
- ❸ The model is deployed in production
- ❹ The model is evaluated on test data

ANSWER: B

DETAILED EXPLANATION:

- Trained on unlabeled text
- Understands word relationships
- Learns language patterns
- Foundation for fine-tuning

WHY OTHER OPTIONS ARE WRONG:

- **A:** Fine-tuning happens after pre-training
- **C:** Deployment happens after training
- **D:** Evaluation happens after training

EXAM TRAP: Pre-training = **unlabeled text** learning!

QUESTION: What is the purpose of fine-tuning an LLM?

OPTIONS:

- ❶ To start training from scratch
- ❷ To transfer knowledge to specific tasks using labeled data
- ❸ To clean the data
- ❹ To tokenize the text

ANSWER: B

DETAILED EXPLANATION:

- Transfer knowledge from pre-training
- Use labeled data for specific tasks
- Adapt model to domain
- Enable task-specific performance

WHY OTHER OPTIONS ARE WRONG:

- **A:** Fine-tuning uses existing pre-trained model
- **C:** Data cleaning happens before training
- **D:** Tokenization happens before training

EXAM TRAP: Fine-tuning = **task-specific adaptation** with labeled data!

QUESTION: What is RLHF in LLM training?

OPTIONS:

- ① Reinforcement Learning Through Human Feedback
- ② Recurrent Learning with Human Feedback
- ③ Regularization Learning with Human Feedback
- ④ Regression Learning with Human Feedback

ANSWER: A

DETAILED EXPLANATION:

- Reinforcement Learning through Human Feedback
- Used in alignment phase
- Conforms model to human preferences
- Based on user feedback

WHY OTHER OPTIONS ARE WRONG:

- **B:** Incorrect expansion of RLHF
- **C:** Incorrect expansion of RLHF
- **D:** Incorrect expansion of RLHF

EXAM TRAP: RLHF = Reinforcement Learning through Human Feedback!

QUESTION: Why are most LLMs cloud-based?

OPTIONS:

- ❶ They are too small to store locally
- ❷ They are huge (billions of parameters) and training is only done once
- ❸ It's cheaper to store locally
- ❹ They don't require storage

ANSWER: B

DETAILED EXPLANATION:

- Weight matrices are massive (billions of parameters)
- Infeasible to store locally
- Training only done once (too expensive to repeat)
- Reside remotely on cloud storage

WHY OTHER OPTIONS ARE WRONG:

- **A:** They are huge, not small
- **C:** Local storage is not feasible
- **D:** They need storage (cloud storage)

EXAM TRAP: LLMs are **cloud-based** due to **size**!

QUESTION: Which GPT generation introduced "no-shot fine-tuning"?

OPTIONS:

- ❶ GPT-1
- ❷ GPT-2
- ❸ GPT-3
- ❹ GPT-4

ANSWER: B

DETAILED EXPLANATION:

- GPT-2 removed need for second stage fine-tuning
- No-shot fine-tuning became possible
- Model adapts to new situations without examples
- Trained on larger, more varied corpus

WHY OTHER OPTIONS ARE WRONG:

- **A:** GPT-1 required fine-tuning
- **C:** GPT-3 improved but GPT-2 introduced no-shot
- **D:** GPT-4 came later

EXAM TRAP: GPT-2 introduced no-shot fine-tuning!

QUESTION: What is no-shot fine-tuning in LLMs?

OPTIONS:

- ❶ Requires many examples to adapt
- ❷ Adapts to new situations without any examples
- ❸ Requires exactly one example
- ❹ Cannot adapt to new situations

ANSWER: B

DETAILED EXPLANATION:

- No supervised examples needed
- Model adapts to new tasks
- Relies on pre-training knowledge
- GPT-2 enabled this

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is few-shot, not no-shot
- **C:** This is one-shot, not no-shot
- **D:** No-shot enables adaptation

EXAM TRAP: No-shot = no examples needed!

QUESTION: Which type of chatbot is most advanced?

OPTIONS:

- ❶ FAQ-based chatbots
- ❷ Flow-based chatbots
- ❸ LLM-powered chatbots
- ❹ All are equally advanced

ANSWER: C

DETAILED EXPLANATION:

- FAQ-based: Standard responses, limited
- Flow-based: Interactive, multi-step
- LLM-powered: Deep understanding, human-like
- LLM-powered chatbots are the most advanced

WHY OTHER OPTIONS ARE WRONG:

- **A:** FAQ-based are the simplest
- **B:** Flow-based are intermediate
- **D:** They have different levels of sophistication

EXAM TRAP: LLM-powered chatbots are **most advanced!**

QUESTION: What is prompt engineering?

OPTIONS:

- ❶ Training the model from scratch
- ❷ The art and science of designing prompts for effective LLM use
- ❸ Evaluating model performance
- ❹ Tokenizing text

ANSWER: B

DETAILED EXPLANATION:

- Designing prompts for best results
- Iterative process
- Provide context and examples
- Guide model responses

WHY OTHER OPTIONS ARE WRONG:

- **A:** Training is different from prompting
- **C:** Evaluation is different from prompting
- **D:** Tokenization is a pre-processing step

EXAM TRAP: Prompt engineering = **designing effective prompts!**

QUESTION: What is zero-shot prompting?

OPTIONS:

- ❶ Providing many examples before the question
- ❷ Providing no examples before the question
- ❸ Providing one example before the question
- ❹ Providing multiple examples with reasoning

ANSWER: B

DETAILED EXPLANATION:

- No examples provided
- User inputs question directly
- Most common type
- "What is the capital of France?"

WHY OTHER OPTIONS ARE WRONG:

- A: This is few-shot prompting
- C: This is one-shot prompting
- D: This is few-shot CoT

EXAM TRAP: Zero-shot = no examples!

QUESTION: What is few-shot prompting?

OPTIONS:

- ❶ Providing no examples
- ❷ Providing one example
- ❸ Providing multiple examples before the target question
- ❹ Providing examples after the answer

ANSWER: C

DETAILED EXPLANATION:

- Provides multiple examples in the prompt
- Gives the model more context
- Helps produce more accurate and consistent responses
- Example: Several Q&A pairs are provided before the target question

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is zero-shot prompting
- **B:** This is one-shot prompting
- **D:** Examples are provided before, not after, the target question

EXAM TRAP: Few-shot = multiple examples before the target!

QUESTION: What is Chain of Thought (CoT) prompting?

OPTIONS:

- ❶ Prompting with the final answer only
- ❷ Prompting with intermediate reasoning steps
- ❸ Prompting with no text
- ❹ Prompting with random words

ANSWER: B

DETAILED EXPLANATION:

- Include reasoning steps in prompt
- Solves complex reasoning tasks
- Improves multi-step reasoning
- "Let's think step by step"

WHY OTHER OPTIONS ARE WRONG:

- **A:** Standard prompting gives final answer only
- **C:** CoT requires text
- **D:** CoT uses structured reasoning

EXAM TRAP: CoT = intermediate reasoning steps!

QUESTION: What is context length in LLMs?

OPTIONS:

- ❶ The number of parameters in the model
- ❷ The maximum number of tokens a model can process in a single sequence
- ❸ The number of layers in the model
- ❹ The training time of the model

ANSWER: B

DETAILED EXPLANATION:

- Maximum tokens in input/output sequence
- Also called context window
- Limits model's working memory
- Important for long documents

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is parameter count
- **C:** This is model depth
- **D:** This is training duration

EXAM TRAP: Context length = **tokens model can process!**

QUESTION: What does statelessness mean in LLMs?

OPTIONS:

- ❶ LLMs remember all past conversations
- ❷ LLMs don't retain information between prompts
- ❸ LLMs retain information for one day
- ❹ LLMs have perfect memory

ANSWER: B

DETAILED EXPLANATION:

- Each prompt processed independently
- No memory of previous conversations
- By design
- Users must feed previous responses for context

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is stateful (opposite)
- **C:** There is no time-based retention
- **D:** LLMs have no perfect memory

EXAM TRAP: Stateless = **no memory between prompts!**

QUESTION: What does the temperature parameter control in LLMs?

OPTIONS:

- ❶ The number of tokens generated
- ❷ The randomness/creativity of output
- ❸ The length of the output
- ❹ The speed of generation

ANSWER: B

DETAILED EXPLANATION:

- Controls randomness/creativity
- Shapes probability distribution
- Lower = more predictable, deterministic
- Higher = more creative, varied

WHY OTHER OPTIONS ARE WRONG:

- **A:** Token count is a separate parameter
- **C:** Output length is a separate parameter
- **D:** Speed is determined by hardware

EXAM TRAP: Temperature controls **randomness/creativity!**

QUESTION: What does Top-K sampling do in LLMs?

OPTIONS:

- ❶ Considers all tokens equally
- ❷ Only considers the K most probable tokens
- ❸ Randomly selects tokens
- ❹ Always selects the most probable token

ANSWER: B

DETAILED EXPLANATION:

- Restricts to K most probable tokens
- Low K = less creativity
- High K = more variety
- Example: K=3 means only top 3 considered

WHY OTHER OPTIONS ARE WRONG:

- **A:** Top-K restricts, doesn't consider all
- **C:** Top-K is deterministic, not random
- **D:** This is temperature=0, not Top-K

EXAM TRAP: Top-K = **only K most probable tokens!**

QUESTION: What is Agentic AI?

OPTIONS:

- ❶ AI that only generates text
- ❷ AI agents that take inputs and perform actions autonomously
- ❸ AI that only classifies data
- ❹ AI that requires constant human supervision

ANSWER: B

DETAILED EXPLANATION:

- Next frontier in AI
- Autonomous action
- Uses LLM outputs
- Dynamic decision-making

WHY OTHER OPTIONS ARE WRONG:

- **A:** Agentic AI goes beyond text generation
- **C:** Agentic AI takes action, not just classify
- **D:** Agentic AI operates with limited supervision

EXAM TRAP: Agentic AI = **autonomous action**!

QUESTION: Why is GenAI evaluation more complex than traditional ML?

OPTIONS:

- ❶ GenAI outputs are always correct
- ❷ GenAI outputs are open-ended and unstructured
- ❸ GenAI outputs are always deterministic
- ❹ GenAI outputs are simple to evaluate

ANSWER: B

DETAILED EXPLANATION:

- Unstructured responses
- No single correct answer
- Many valid variations
- Hallucination risk

WHY OTHER OPTIONS ARE WRONG:

- **A:** GenAI outputs can be incorrect
- **C:** GenAI outputs are probabilistic
- **D:** Evaluation is more complex, not simple

EXAM TRAP: GenAI evaluation is **complex** due to open-ended outputs!

QUESTION: What is hallucination in GenAI?

OPTIONS:

- ❶ Output that is always correct
- ❷ Output that appears reasonable but is factually incorrect
- ❸ Output that is completely nonsensical
- ❹ Output that is copied verbatim from training data

ANSWER: B

DETAILED EXPLANATION:

- Plausible but incorrect
- Not supported by input
- Caused by probabilistic generation
- A serious GenAI risk

WHY OTHER OPTIONS ARE WRONG:

- **A:** Hallucinations are factually incorrect
- **C:** Hallucinations appear reasonable, not nonsensical
- **D:** Hallucinations are not necessarily copied

EXAM TRAP: Hallucination = **reasonable but incorrect** output!

QUESTION: What is a primary cause of hallucination in LLMs?

OPTIONS:

- ❶ Perfect training data
- ❷ Probabilistic token selection
- ❸ Deterministic outputs
- ❹ Small model size

ANSWER: B

DETAILED EXPLANATION:

- Tokens chosen based on probabilities
- Higher temperature = more variation
- May pick incorrect tokens
- Knowledge gaps filled with plausible answers

WHY OTHER OPTIONS ARE WRONG:

- **A:** Training data is never perfect
- **C:** LLM outputs are probabilistic, not deterministic
- **D:** Larger models can also hallucinate

EXAM TRAP: Hallucination = probabilistic nature + knowledge gaps!

QUESTION: What does robustness in GenAI evaluation measure?

OPTIONS:

- ❶ How fast the model runs
- ❷ How well the model performs on novel/unseen inputs
- ❸ How many parameters the model has
- ❹ How much data the model was trained on

ANSWER: B

DETAILED EXPLANATION:

- Out of Distribution (OOD) performance
- Adversarial robustness
- Jailbreak resistance
- Prompt sensitivity

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is operational performance
- **C:** This is model size, not robustness
- **D:** This is training data size

EXAM TRAP: Robustness = performance on unseen/attacked inputs!

QUESTION: What is WEAT used to measure?

OPTIONS:

- ❶ Model accuracy
- ❷ Bias using cosine similarity between sets of words
- ❸ Model speed
- ❹ Model size

ANSWER: B

DETAILED EXPLANATION:

- WEAT = Word Embedding Association Test
- Measures bias using cosine similarity
- Between sets of words
- Normalized score

WHY OTHER OPTIONS ARE WRONG:

- **A:** WEAT measures bias, not accuracy
- **C:** WEAT doesn't measure speed
- **D:** WEAT doesn't measure size

EXAM TRAP: WEAT = **bias measurement** with cosine similarity!

QUESTION: What is "Time to First Token" in LLM evaluation?

OPTIONS:

- ❶ Total generation time
- ❷ Time between request and first token generation
- ❸ Time between tokens
- ❹ Time to complete the entire response

ANSWER: B

DETAILED EXPLANATION:

- Perceived responsiveness
- Important for user experience
- Measures initial delay
- Affects user satisfaction

WHY OTHER OPTIONS ARE WRONG:

- **A:** Total generation time is different
- **C:** This is time between tokens
- **D:** This is total time, not first token

EXAM TRAP: Time to First Token = **initial response delay!**

QUESTION: What does BLEU score measure in text generation?

OPTIONS:

- ❶ Creativity of output
- ❷ n-gram precision compared to reference text
- ❸ Speed of generation
- ❹ Model size

ANSWER: B

DETAILED EXPLANATION:

- BLEU = Bilingual Evaluation Understudy
- n-gram precision
- Adds brevity penalty
- 0-1 scale

WHY OTHER OPTIONS ARE WRONG:

- **A:** BLEU measures similarity, not creativity
- **C:** BLEU doesn't measure speed
- **D:** BLEU doesn't measure model size

EXAM TRAP: BLEU = n-gram precision vs reference!

QUESTION: What is ROUGE used for in GenAI evaluation?

OPTIONS:

- ❶ Evaluating image quality
- ❷ Evaluating summaries/translations against reference
- ❸ Evaluating audio quality
- ❹ Evaluating model speed

ANSWER: B

DETAILED EXPLANATION:

- ROUGE = Recall-Oriented Understudy
- For summaries and translations
- n-gram matching
- 0-1 scale

WHY OTHER OPTIONS ARE WRONG:

- **A:** ROUGE is for text, not images
- **C:** ROUGE is for text, not audio
- **D:** ROUGE measures quality, not speed

EXAM TRAP: ROUGE = [summary/translation evaluation](#)!

QUESTION: What does FID measure in image generation?

OPTIONS:

- ❶ How fast images are generated
- ❷ How close generated images are to real images
- ❸ How large images are
- ❹ How many colors images have

ANSWER: B

DETAILED EXPLANATION:

- FID = Fréchet Inception Distance
- Compares generated vs real images
- Distribution parameters (mean, covariance)
- Lower FID = better quality

WHY OTHER OPTIONS ARE WRONG:

- **A:** FID measures quality, not speed
- **C:** FID doesn't measure image size
- **D:** FID doesn't measure color count

EXAM TRAP: FID = **generated vs real image comparison!**

QUESTION: What does MMLU benchmark evaluate?

OPTIONS:

- ❶ Image generation quality
- ❷ Language understanding across 57 subjects
- ❸ Audio quality
- ❹ Model speed

ANSWER: B

DETAILED EXPLANATION:

- MMLU = Massive Multitask Language Understanding
- Evaluates across 57 subjects
- Humanities, social sciences, STEM
- Multiple-choice format

WHY OTHER OPTIONS ARE WRONG:

- **A:** MMLU is text-based, not image
- **C:** MMLU is text-based, not audio
- **D:** MMLU measures knowledge, not speed

EXAM TRAP: MMLU = multitask language understanding across 57 subjects!

QUESTION: What is data leakage in LLM benchmarking?

OPTIONS:

- ❶ Benchmark data being included in training data
- ❷ Perfect model performance
- ❸ Model not being tested
- ❹ Model being too slow

ANSWER: A

DETAILED EXPLANATION:

- Benchmark data in training set
- Model "memorizes" answers
- Inflated performance
- False indication of ability

WHY OTHER OPTIONS ARE WRONG:

- **B:** Data leakage leads to false performance
- **C:** Data leakage means model IS tested (incorrectly)
- **D:** Data leakage is about data contamination, not speed

EXAM TRAP: Data leakage = **benchmark data in training!**

QUESTION: What is benchmark saturation?

OPTIONS:

- ❶ Models cannot solve the benchmark
- ❷ Models reach maximum scores, making differentiation impossible
- ❸ The benchmark is too difficult
- ❹ The benchmark is not used

ANSWER: B

DETAILED EXPLANATION:

- Models reach ceiling scores
- Cannot differentiate performance
- Benchmarks become outdated
- Need new challenges

WHY OTHER OPTIONS ARE WRONG:

- **A:** Saturation means models CAN solve it
- **C:** Saturation means it's too easy, not too difficult
- **D:** Saturation means it's used but ineffective

EXAM TRAP: Saturation = models max out benchmark!

QUESTION: What is cherry picking in LLM benchmark reporting?

OPTIONS:

- ❶ Reporting all benchmark results fairly
- ❷ Only highlighting good benchmark results, omitting weaker ones
- ❸ Using the latest benchmarks
- ❹ Comparing models equally

ANSWER: B

DETAILED EXPLANATION:

- Selectively report good results
- Omit weaker benchmarks
- Misleading representation
- Selective reporting

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is fair reporting, not cherry picking
- **C:** Cherry picking is about selection, not recency
- **D:** Cherry picking is unfair comparison

EXAM TRAP: Cherry picking = **selective reporting** of good results!

QUESTION: What does Hallucination Rate measure?

OPTIONS:

- ❶ Model speed
- ❷ Model's propensity for inaccurate output
- ❸ Model size
- ❹ Model training time

ANSWER: B

DETAILED EXPLANATION:

- Quantifies hallucination frequency
- Essential risk metric
- Identifies unreliable outputs
- Critical for risk management

WHY OTHER OPTIONS ARE WRONG:

- **A:** Hallucination rate is about quality, not speed
- **C:** Hallucination rate is about accuracy, not size
- **D:** Hallucination rate is about output quality, not training

EXAM TRAP: Hallucination Rate = **propensity for inaccurate output!**

QUESTION: What are key operational performance metrics for LLMs?

OPTIONS:

- ❶ Accuracy and precision only
- ❷ Latency, throughput, and token efficiency
- ❸ Model size only
- ❹ Training data size only

ANSWER: B

DETAILED EXPLANATION:

- **Latency:** Response time
- **Throughput:** Queries per time
- **Token efficiency:** Quality per token
- Critical for production deployment

WHY OTHER OPTIONS ARE WRONG:

- **A:** Accuracy/precision are quality metrics
- **C:** Model size is a characteristic, not performance
- **D:** Training data size is a characteristic

EXAM TRAP: Operational metrics = **latency, throughput, token efficiency!**

QUESTION: What is a common text processing application of GenAI?

OPTIONS:

- ❶ Image generation
- ❷ Document summarization
- ❸ Video creation
- ❹ Audio synthesis

ANSWER: B

DETAILED EXPLANATION:

- Report generation
- Document summarization
- Legal analysis support
- Writing assistance

WHY OTHER OPTIONS ARE WRONG:

- **A:** Image generation is a different modality
- **C:** Video creation is a different modality
- **D:** Audio synthesis is a different modality

EXAM TRAP: Text processing includes **summarization** and **report generation**!

QUESTION: How can GenAI help with compliance monitoring?

OPTIONS:

- ❶ By ignoring all regulations
- ❷ By analyzing SEC filings and detecting emerging risks
- ❸ By removing all compliance requirements
- ❹ By creating new regulations

ANSWER: B

DETAILED EXPLANATION:

- Analyze SEC filings
- Review company reports
- Detect emerging risks
- Regulatory compliance

WHY OTHER OPTIONS ARE WRONG:

- **A:** Compliance requires attention to regulations
- **C:** Requirements cannot be ignored
- **D:** GenAI doesn't create regulations

EXAM TRAP: GenAI helps with **compliance monitoring**!

QUESTION: What are examples of finance-specialized LLMs?

OPTIONS:

- ❶ ChatGPT and Claude
- ❷ Bloomberg GPT and FinGPT
- ❸ DALL-E and Stable Diffusion
- ❹ Sora and Gemini

ANSWER: B

DETAILED EXPLANATION:

- Bloomberg GPT
- FinGPT
- Financial news analysis
- Trading signal generation

WHY OTHER OPTIONS ARE WRONG:

- **A:** These are general-purpose LLMs
- **C:** These are image generation models
- **D:** These are multimodal/general models

EXAM TRAP: Finance-specific LLMs = Bloomberg GPT, FinGPT!

QUESTION: Which model is specialized for code generation?

OPTIONS:

- ❶ Codex (OpenAI)
- ❷ DALL-E
- ❸ Sora
- ❹ Whisper

ANSWER: A

DETAILED EXPLANATION:

- Codex by OpenAI
- Code generation from descriptions
- Natural language to code
- GitHub Copilot uses Codex

WHY OTHER OPTIONS ARE WRONG:

- **B:** DALL-E generates images
- **C:** Sora generates videos
- **D:** Whisper converts speech to text

EXAM TRAP: Codex = **code generation** from OpenAI!

QUESTION: What is GitHub Copilot?

OPTIONS:

- ❶ An image generation tool
- ❷ An AI pair programmer for code suggestions
- ❸ A speech recognition tool
- ❹ A video generation tool

ANSWER: B

DETAILED EXPLANATION:

- AI pair programmer
- Code suggestions
- Based on Codex/OpenAI
- Integrated with GitHub

WHY OTHER OPTIONS ARE WRONG:

- **A:** Image generation is different
- **C:** Speech recognition is different
- **D:** Video generation is different

EXAM TRAP: GitHub Copilot = AI pair programmer!

QUESTION: How does GenAI help with fraud detection?

OPTIONS:

- ❶ By ignoring transactions
- ❷ By automating monitoring and identifying anomalies
- ❸ By removing all alerts
- ❹ By approving all transactions

ANSWER: B

DETAILED EXPLANATION:

- Automate monitoring
- Compile transaction data
- Identify anomalies/outliers
- Real-time detection

WHY OTHER OPTIONS ARE WRONG:

- **A:** Ignoring transactions would miss fraud
- **C:** Removing alerts misses fraud
- **D:** Approving all transactions allows fraud

EXAM TRAP: GenAI = automated fraud detection!

QUESTION: What is Mastercard's Decision Intelligence Pro?

OPTIONS:

- ❶ A general-purpose chatbot
- ❷ A dedicated GenAI fraud detection tool
- ❸ A video generation tool
- ❹ A speech recognition tool

ANSWER: B

DETAILED EXPLANATION:

- Dedicated GenAI fraud detection
- Advanced pattern recognition
- Financial transaction monitoring
- Real-time detection

WHY OTHER OPTIONS ARE WRONG:

- **A:** It's specialized, not general-purpose
- **C:** It's for fraud detection, not video
- **D:** It's for fraud detection, not speech

EXAM TRAP: Decision Intelligence Pro = fraud detection tool!

QUESTION: How do LLMs help with cybersecurity?

OPTIONS:

- ❶ By ignoring all threats
- ❷ By monitoring for potential threats and identifying vulnerabilities
- ❸ By removing all security measures
- ❹ By approving all access requests

ANSWER: B

DETAILED EXPLANATION:

- Monitor emails and data streams
- Identify potential cyber threats
- Cross-reference with system profile
- Pinpoint vulnerabilities

WHY OTHER OPTIONS ARE WRONG:

- **A:** Ignoring threats is dangerous
- **C:** Security measures are essential
- **D:** Approving all requests allows attacks

EXAM TRAP: LLMs = cybersecurity monitoring and threat detection!

QUESTION: What does Cisco Systems use LLMs for?

OPTIONS:

- ❶ Image generation
- ❷ Malware detection
- ❸ Speech recognition
- ❹ Video creation

ANSWER: B

DETAILED EXPLANATION:

- Custom LLM for malware detection
- Security analysis
- Threat identification
- Cybersecurity protection

WHY OTHER OPTIONS ARE WRONG:

- **A:** Cisco focuses on cybersecurity, not images
- **C:** Cisco focuses on malware, not speech
- **D:** Cisco focuses on detection, not video

EXAM TRAP: Cisco = **malware detection** with LLMs!

QUESTION: What is CrowdStrike's Charlotte AI used for?

OPTIONS:

- ❶ Image generation
- ❷ Cybersecurity and threat response
- ❸ Speech recognition
- ❹ Video creation

ANSWER: B

DETAILED EXPLANATION:

- Charlotte AI for cybersecurity
- Automated threat response
- Security operations
- Pattern recognition

WHY OTHER OPTIONS ARE WRONG:

- **A:** Charlotte AI is for security, not images
- **C:** Charlotte AI is for threat detection, not speech
- **D:** Charlotte AI is for security, not video

EXAM TRAP: Charlotte AI = cybersecurity threat response!

QUESTION: What is the EXL Insurance LLM used for?

OPTIONS:

- ❶ General purpose chat
- ❷ Claims processing, underwriting, and customer service
- ❸ Image generation
- ❹ Video creation

ANSWER: B

DETAILED EXPLANATION:

- Claims processing
- Underwriting
- Customer service
- Insurance-specific tasks

WHY OTHER OPTIONS ARE WRONG:

- **A:** EXL is insurance-specific, not general
- **C:** EXL is text-based, not image
- **D:** EXL is document processing, not video

EXAM TRAP: EXL Insurance LLM = claims + underwriting + customer service!

QUESTION: What performance improvement did EXL Insurance LLM achieve?

OPTIONS:

- ❶ 10% reduction in operational costs
- ❷ 30% reduction in operational costs
- ❸ 50% reduction in operational costs
- ❹ No reduction in operational costs

ANSWER: B

DETAILED EXPLANATION:

- 30% reduction in operational costs
- Through efficient claims processing
- Streamlined underwriting
- Automated customer service

WHY OTHER OPTIONS ARE WRONG:

- **A:** Too low compared to actual results
- **C:** Too high compared to actual results
- **D:** There was a significant reduction

EXAM TRAP: EXL Insurance LLM = 30% cost reduction!

QUESTION: Which investment firms are using LLMs?

OPTIONS:

- ❶ Only BlackRock
- ❷ BlackRock, Bridgewater, and AQR
- ❸ Only Bridgewater
- ❹ No investment firms use LLMs

ANSWER: B

DETAILED EXPLANATION:

- **BlackRock:** LLMs in investment process
- **Bridgewater:** \$2 billion ML fund
- **AQR:** LLM integration for strategies

WHY OTHER OPTIONS ARE WRONG:

- **A:** More firms than just BlackRock
- **C:** More firms than just Bridgewater
- **D:** Many firms use LLMs

EXAM TRAP: BlackRock, Bridgewater, AQR = **all using LLMs!**

QUESTION: What areas of finance are being transformed by GenAI?

OPTIONS:

- ❶ Only fraud detection
- ❷ Fraud detection, customer service, document processing, and investment
- ❸ Only customer service
- ❹ Only document processing

ANSWER: B

DETAILED EXPLANATION:

- **Fraud Detection:** Automated monitoring
- **Customer Service:** Chatbots
- **Document Processing:** Summarization
- **Investment:** Market analysis and trading

WHY OTHER OPTIONS ARE WRONG:

- **A:** GenAI impacts multiple areas
- **C:** GenAI impacts more than customer service
- **D:** GenAI impacts more than document processing

EXAM TRAP: GenAI transforms **multiple financial sectors!**

QUESTION: What is the most comprehensive summary of Generative AI?

OPTIONS:

- ❶ GenAI is only about text generation
- ❷ GenAI creates new content, uses Word2Vec/RNNs/Transformers/LLMs, requires careful evaluation, and has broad applications
- ❸ GenAI is only about image generation
- ❹ GenAI doesn't require evaluation

ANSWER: B

DETAILED EXPLANATION:

- **Creates new content:** Text, images, audio, video
- **Key technologies:** Word2Vec, RNNs, Transformers, LLMs
- **Evaluation:** Quality, hallucination, robustness, fairness
- **Applications:** Broad across all financial sectors

WHY OTHER OPTIONS ARE WRONG:

- **A:** GenAI includes more than text
- **C:** GenAI includes more than images
- **D:** Evaluation is essential for GenAI

EXAM SUCCESS: Understand **technology**, **evaluation**, and **applications**!